

Supplementary Information

Typed material-flow ledgers for componentwise sustainability diagnostics

Amin Abaee

Independent Researcher

ORCID: 0000-0002-0019-1842

amin_abaee@ut.ac.ir

19 September 2026

Abstract

This Supplementary Information formalizes the certification layers of the typed material-flow ledger, specifies the interface between the closed ledger and institutional delay dynamics, reports the application classifications, and defines operational weak- and strong-sustainability regimes. The formalization is conditional on declared types, routes, barriers, demand and horizon; it does not constitute a universal theorem about sustainability ethics or an empirical calibration of any named domain.

Scope

The Supplementary Information contains four components:

1. the extended inventory of accounting, positivity, barrier, depletion and aggregation results;
2. the exact shared object and the non-reduction boundary for institutional delay dynamics;
3. full tables for the groundwater, phosphate and fisheries records, including status and vintage information; and
4. an operational weak/strong-sustainability formalization that states the admissible closure assumptions.

These components use the definitions and evidence classifications of the article. Procedural details are given in the methods companion (Abaee, 2026a), and the accounting-standards interpretation is given in the accompanying commentary (Abaee, 2026b). The formalization is conditional on the declarations stated here; it does not promote an interpretation to an unrestricted theorem.

Every application number remains a bounded descriptive record. The groundwater index is record-relative; the phosphate quotient is an economic reserve-class ratio; and the fisheries quantity is a removals-only pressure time. None is presented here as a calibrated forecast.

S0. Notation and the declaration discipline

A typed ledger has state $x(t) \in \mathbb{R}_+^n$, non-negative primitive fluxes $v(t) \in \mathbb{R}_+^J$, a declared boundary schedule $b(t)$, a type structure $\mathcal{T}$, and an incidence operator $S_{\mathcal{T}}$. Its balance is

$$\dot{x} = S_{\mathcal{T}}v + b. \quad (1)$$

A composition matrix C maps compartments to declared conserved substance classes, $S = Cx$. A service is a readout $O(x, u, \theta)$, not an additional conserved mass compartment. A margin or barrier is represented by an affine map $m(x) = Gx + a \geq 0$ when that form is appropriate.

A primitive column is a *transfer* when its non-zero entries are signed unit entries within one declared type and unit class. It is a *conversion* when the source and product entries carry a declared stoichiometric coefficient and a routed destination. A cross-type sum is not admitted without that conversion declaration. A donor-limited primitive outflow satisfies

$$v_j(t) = 0 \quad \text{whenever its donor compartment is empty.} \quad (2)$$

Every yield below one must route its residual to a represented compartment or an explicitly declared boundary term. These are declaration rules; they are not inferred from a diagnostic label such as “reserve”, “waste” or “unsustainable portion”.

The status vocabulary is four-valued. **established** means that the declared obligation has a witness; **not established** means that the obligation is stated but not discharged by the declared record; **not applicable** means that the object has no live obligation of that kind; and **refuted** requires a witness of the negation. A missing trajectory must not be reported as a refutation.

S1. Extended theorem inventory

Table 1 gives the extended inventory. The labels are local to this Supplementary Information and identify the results stated below. Proof sketches are supplied where the argument is short.

The certification stack is deliberately layered:

$$\begin{aligned} \text{typed} &\longrightarrow \text{balanced} \longrightarrow \text{conserved} \longrightarrow \text{admissible} \longrightarrow \text{safe}, \\ &\quad \text{safe} \longrightarrow \text{closure-certified}. \end{aligned} \quad (3)$$

The arrows name proof obligations, not automatic logical implications. “Typed” checks units and routes; “balanced” checks the displayed equations; “conserved” checks the declared moiety law; “admissible” checks non-negativity and donor limits; “safe” checks componentwise barriers on a horizon; and “closure-certified” additionally requires an admissible return/substitution witness for the declared service. A positive aggregate cannot skip from a lower layer to a higher one.

Table 1: Extended theorem inventory and scope.

Result	Statement supplied to the JIE reader	Boundary of the result
S1.1 Flux reconstruction	Integrating (1) reconstructs $x(t)$ from the initial state, typed fluxes and declared boundary terms.	An identity for the declared ledger; it does not identify unmeasured fluxes without additional data or a programme.
S1.2 Conservation reduction	If $\ell^\top CS_{\mathcal{T}} = 0$, then $\ell^\top Cx(T) = \ell^\top Cx(0) + \int_0^T \ell^\top Cb(t) dt$.	Boundary and residual terms remain visible; an undeclared residual is not silently zero.
S1.2a Double-count guard	An aggregate row and the disjoint rows that sum to it are mutually exclusive inputs to a total; a derived readout is not added again as a primitive flow.	A data-selection audit is required; overlapping rows can change totals and ratios.
S1.3 Flux envelope	Componentwise flux boxes produce lower and upper moiety envelopes by positive/negative matrix parts.	A box is conservative and may contain jointly unrealizable flux choices; it is not a forecast.
S1.4 Barrier corollary	If the envelope lies between declared barriers for the full horizon, every trajectory in the box is barrier-safe.	Safety is conditional on the box, the horizon and the declared barriers.
S1.5 Uniform-drift bound	If $\dot{S} \leq -\varepsilon < 0$ above a barrier B , then $\tau_B \leq (S(0) - B)/\varepsilon$.	The uniform margin is the substantive assumption; positive extraction alone is insufficient.
S1.6 Critical-margin budget	A non-negative multiplier satisfying $\lambda^\top GS_{\mathcal{T}} + c^\top \leq 0$ bounds cumulative service by initial margin plus declared boundary support.	$V > 0$ is not a componentwise certificate, and multiplier infeasibility is not proof of safety.
S1.7 Closed finite-donor mass	For the closed natural block, $M = N + A^{\text{act}} + A^{\text{geo}} + U$ obeys $\dot{M} = -qEN - C^{A,\text{lim}}$.	The identity is for the declared routing; internal transfers do not exhaust a conserved moiety.
S1.8 Orthant invariance	Donor limitation gives forward invariance of $\mathbb{R}_+^n$ by a face-by-face tangent-cone argument.	It proves non-negativity, not barrier safety, closure, or sustainability.
S1.9 No interior rest	With positive extraction effort and the stated constitutive laws, the closed finite-donor block has no interior positive-flux rest point.	The claim is for the closed block and its hypotheses, not for an open target-relaxation completion.

Result	Statement supplied to the JIE reader	Boundary of the result
S1.10 Rest-set and budget results	The zero-extraction rest set includes extinction, carrying-capacity and frozen-support faces; extraction is L^1 over a finite donor budget.	These are structural results, not empirical identification of groundwater, phosphate or fisheries.
S1.11 No weighted certification	A non-negative weighted sum does not imply componentwise adequacy on a feasible domain containing a positive aggregate with a negative component.	An additional domain theorem could change the implication; absent one, scalar communication is not certification.
S1.12 Depletion is compartmental	A finite crossing concerns a declared compartment, barrier or readout, not total conserved mass.	A reserve-class quotient and a record-relative anomaly index are not automatically physical hitting times.
S1.13 First-passage surrogates	Brownian and geometric-Brownian first-passage formulae are valid for their declared surrogate processes.	The surrogates do not stochastically complete the typed ledger and do not identify a physical failure threshold.
S1.14 Delay-interface non-reduction	The closed ledger and open working delay system share a decline identity under a specialization but do not share a dynamic reduction.	Shared diagnostics do not transfer Hopf, equilibrium or periodic-orbit results.

S1.1 Balance, conservation and flux envelopes

Integrating (1) gives the reconstruction identity

$$x(T) - x(0) = \int_0^T S_{\mathcal{T}} v(t) dt + \int_0^T b(t) dt. \quad (4)$$

If $\ell^\top C S_{\mathcal{T}} = 0$, multiplication by $\ell^\top C$ gives

$$\ell^\top C x(T) = \ell^\top C x(0) + \int_0^T \ell^\top C b(t) dt. \quad (5)$$

With a declared residual d_x , the right-hand side includes $\int_0^T \ell^\top C d_x(t) dt$ and its magnitude is bounded only by the declared residual budget $\varepsilon_\ell(T)$. If that budget is missing, the conservation status is **not established**, not established at zero.

Proposition S1.2a (Double-counting safeguard). Let q_0 be an aggregate record and let $q_1, \dots, q_k$ be mutually exclusive component records with $q_0 = \sum_i q_i$. A total must use either q_0 or $\sum_i q_i$, not $q_0 + \sum_i q_i$. The same selector rule applies when a derived readout is computed from a primitive flow: the readout is not added as a second flow.

Proof. The mixed total is $q_0 + \sum_i q_i = 2q_0$ whenever the components exhaust the aggregate and $q_0 > 0$. For a ratio, the numerator and denominator can be inflated by different overlapping scopes, so the quotient need not be invariant. Choosing one mutually exclusive aggregation level and recording the row selector prevents the duplication. The safeguard is bookkeeping, not a substantive conservation theorem.

Suppose $v(t) \in [\underline{v}(t), \bar{v}(t)]$ and $b(t) \in [\underline{b}(t), \bar{b}(t)]$. Write $A^+ = \max(A, 0)$ and $A^- = \max(-A, 0)$ entrywise. For moiety row m define

$$\varphi_m(t) = (CS_{\mathcal{T}})_m^+ \underline{v}(t) - (CS_{\mathcal{T}})_m^- \bar{v}(t) + C_m^+ \underline{b}(t) - C_m^- \bar{b}(t), \quad (6)$$

$$\psi_m(t) = (CS_{\mathcal{T}})_m^+ \bar{v}(t) - (CS_{\mathcal{T}})_m^- \underline{v}(t) + C_m^+ \bar{b}(t) - C_m^- \underline{b}(t), \quad (7)$$

$$\underline{S}_m(t) = S_m(0) + \int_0^t \varphi_m(\tau) d\tau, \quad \bar{S}_m(t) = S_m(0) + \int_0^t \psi_m(\tau) d\tau. \quad (8)$$

Then $\underline{S}_m(t) \leq S_m(t) \leq \bar{S}_m(t)$. The proof is pointwise: the positive and negative coefficient parts attain the extrema of the linear form on the declared box, and integration preserves the inequality. If $\underline{S}_m(t) \geq \underline{B}_m(t)$ and $\bar{S}_m(t) \leq \bar{B}_m(t)$ for every m, t , the box is a barrier certificate. The certificate is conservative whenever the box includes flux combinations that the coupled dynamics cannot jointly realize.

S1.2 The closed finite-donor block

Let

$$s = \frac{A^{\text{act}}}{A^{\text{act}} + A_0}, \quad \sigma = \frac{A^{\text{geo}}}{A^{\text{geo}} + A_{g0}}, \quad R = rN \left(1 - \frac{N}{K}\right) s, \quad T = \kappa_A N s, \quad (9)$$

and $B = R + T$. The donor primitives are

$$e_{GA} = \omega_A A^{\text{eq, intrinsic}} \sigma, \quad e_{AG} = \omega_A A^{\text{act}}, \quad C^{A, \text{lim}} = C^A \sigma, \quad \text{detritus return} = \gamma_U U. \quad (10)$$

Under the institutional-failure specialization, the closed block is

$$\dot{N} = R - qEN, \quad (11)$$

$$\dot{A}^{\text{act}} = -B + e_{GA} - e_{AG} + \gamma_U U, \quad (12)$$

$$\dot{A}^{\text{geo}} = -e_{GA} + e_{AG}, \quad (13)$$

$$\dot{U} = T - \gamma_U U. \quad (14)$$

Summing gives

$$\dot{M} = -qEN - C^{A, \text{lim}}, \quad M = N + A^{\text{act}} + A^{\text{geo}} + U. \quad (15)$$

The identity separates internal routing from block-boundary export. If $C^A = 0$, the only block loss is harvest; if harvest is routed partly to a product or detritus compartment, the declared block boundary changes accordingly.

At a coordinate face, every negative primitive outflow has the corresponding donor as a factor or is explicitly donor-limited. Thus the vector field lies in the tangent cone of $\mathbb{R}_+^4$ and the closed orthant is forward invariant. At a positive-effort interior rest, $\dot{U} = 0$ and the sum of the active and geological equations forces the regeneration/extraction balance; the living-stock equation requires the opposite sign under positive extraction. The two conditions cannot hold at an interior positive-flux rest under the stated constitutive laws. With extraction integrated over $[0, T]$, (15) yields

$$\int_0^T (qE(t)N(t) + C^{A, \text{lim}}(t)) dt \leq M(0). \quad (16)$$

This is a finite-budget statement, not a forecast of when a named public indicator crosses a threshold.

S1.3 Critical margins and the aggregation obstruction

Let $m(x) = Gx + a \geq 0$, delivered service $y = c^\top v$ with $c \geq 0$, and suppose $\lambda \geq 0$ satisfies

$$\lambda^\top GS_{\mathcal{T}} + c^\top \leq 0 \quad (17)$$

componentwise. Then $V = \lambda^\top m(x)$ satisfies

$$\dot{V} \leq -y + \lambda^\top Gb, \quad \int_0^T y(t) dt \leq V(x(0)) + \int_0^T \lambda^\top Gb(t) dt. \quad (18)$$

The proof is multiplication of the balance by $\lambda^\top G$, followed by $v \geq 0$ and $V(T) \geq 0$. The bound limits the service that can coexist with the declared componentwise margins; it does not make V a replacement for those margins.

For the aggregation obstruction, let $\mathcal{B}$ be a feasible balance domain. If some $b \in \mathcal{B}$ satisfies $b_i < 0$ and $w^\top b > 0$ for $w \geq 0$, then $w^\top b \geq 0$ does not imply $b \geq 0$. The proof is the exhibited counterexample. For $w = (1/2, 1/2)$, $b = (-2, 3)$ gives $w^\top b = 1/2$ while $b_1 < 0$. An aggregate can communicate or rank, but it cannot certify the conjunction of componentwise barriers without an additional implication proved from the physical domain.

S1.4 Depletion is compartmental and first passage is conditional

For a stock or readout S and barrier B , if $\dot{S} \leq -\varepsilon < 0$ above B , then $\tau_B \leq (S(0) - B)/\varepsilon$. The assumption is essential: $\dot{S} = -kS$ has $S(t) > 0$ for every finite t , so the zero barrier is never hit in finite time although a positive barrier is reached at $k^{-1} \log(S(0)/B)$. Internal transfers do not exhaust a conserved moiety. A finite zero or threshold crossing therefore refers to a compartment, a functional readout, an accessibility barrier or an economic classification, not automatically to total mass.

For the Brownian surrogate $A(t) = A_0 + \mu t + \varsigma W_t$ with $\mu < 0$ and record-relative lower barrier $A_{\min}$, the first passage is inverse Gaussian with

$$\nu = \frac{A_0 - A_{\min}}{|\mu|}, \quad \lambda = \frac{(A_0 - A_{\min})^2}{\varsigma^2}, \quad \mathbb{E}[T] = \nu, \quad \text{Var}(T) = \frac{\nu^3}{\lambda}. \quad (19)$$

For the geometric-Brownian comparison $dB_t = -hB_t dt + \varsigma B_t dW_t$, Itô's formula gives a logarithmic drift $h + \varsigma^2/2$ and therefore a corresponding inverse-Gaussian mean. These are conditional statements about declared surrogates. They do not complete the mass ledger stochastically.

S2. Interface with institutional delay dynamics

S2.1 Exact shared object

The closed material ledger comprises the primitive stock-flow equations, typed routing, conservation, positivity, componentwise margins and finite-donor budget. The institutional-delay analysis concerns an open working completion with memory and review delay (Abaee, 2026c). The exact object shared under the single-resource specialization is the decline-pressure identity

$$D(t) = qE(t)N(t) - R(N(t), A(t)) = -\dot{N}(t), \quad \Lambda(t) = [D(t)]_+ = [-\dot{N}(t)]_+. \quad (20)$$

The specialization sets the product, waste and price-response channels and mining intensity to zero, and uses the local stock equation $\dot{N} = R - qEN$. Equation (20) is an interface identity, not a reduction theorem.

A generic retarded institutional completion can be written abstractly as

$$\dot{z}(t) = F(z(t), z_t; \vartheta), \quad z = (N, A^{\text{act}}, A^{\text{geo}}, U, Z, E), \quad (21)$$

$$\dot{Z}(t) = \mathcal{F}(Z_t, \Lambda(t); \vartheta), \quad E(t) = \mathcal{E}(Z_t; \vartheta), \quad (22)$$

where $z_t(\xi) = z(t + \xi)$ for $\xi \in [-\tau, 0]$. This notation identifies the memory input, review delay and effort readout without replacing the equations of the institutional-delay analysis. The interface supplies the typed decline input and its boundary conditions.

S2.2 The two completions

The closed block uses the intrinsic donor-limited target $A^{\text{eq}, \text{intrinsic}}$ and

$$e_{GA} = \omega_A A^{\text{eq}, \text{intrinsic}} \sigma. \quad (23)$$

The open working completion uses the derived target

$$A^{\text{eq}, W} = A^{\text{eq}, \text{intrinsic}} + \frac{\kappa_A K}{\omega_A}, \quad (24)$$

which treats a continuing donor support as an imposed completion. The parameterization uses $A^{\text{eq}, \text{intrinsic}} = 50$, $A^{\text{act}, *} = 397.87$ and $A^{\text{eq}, W} = 5050$.

At the same state, the active-pool vector fields differ by

$$\omega_A (A^{\text{eq}, W} - A^{\text{eq}, \text{intrinsic}} \sigma) - \gamma_U U = \kappa_A K - \gamma_U U \quad (25)$$

under the registered scale separation $\sigma \approx 1$. At the working point, the difference is approximately 0.535 stock units per year; at $U = 0$ the scale is $\kappa_A K = 5.000$. The difference is structural: the working completion omits the detritus return carried by the closed field and uses a derived target.

S2.3 Non-reduction result

Proposition S2.1 (One-way projection). Under the institutional-failure specialization, if the right-hand sides for $(N, A^{\text{act}}, A^{\text{geo}}, U, Z, E)$ contain no excluded macroeconomic state, that block is an exact projection for every parameter value. The converse projection is not available when excluded variables are driven by the block.

Proof sketch. A vector field involving only the displayed block defines an invariant projection; no singular limit or timescale separation is needed. The statement concerns dependency structure only and does not imply that the projected block is the closed physical ledger.

Theorem S2.2 (Non-reduction of the open working completion). There is no exact dynamic reduction, regular perturbation or finite-time tracking correspondence from the closed primitive finite-donor ledger to the open working completion under the registered definitions.

Reason. Five independent obstructions are recorded:

1. the intrinsic and derived targets differ by $\kappa_A K / \omega_A = 5000$ stock units;
2. the two active-pool fields differ at the same state by $\kappa_A K - \gamma_U U$, an $O(1)$ to $O(\kappa_A K)$ term;
3. the working point requires continuing geological support and is not a rest point of the closed finite-donor system;
4. $\varepsilon_G(T) = G_0^{-1} \int_0^T |e_{GA} - e_{AG}| dt$ is a draw diagnostic, not a trajectory-tracking error; and

5. the closed ledger makes sustained extraction L^1 in time, preventing indefinite positive-flux persistence on a finite donor budget.

The permitted relation is a shared diagnostic identity and reconstruction of omitted flows. Hopf crossings, periodic orbits and stability thresholds of the frozen-donor working system do not transfer to the closed ledger. The open working system is therefore treated as a distinct completion with explicit omitted turnover and imposed recharge.

S2.4 Frozen-donor limit and finite-budget reading

With $G = G_0 g$ and $g(0) = 1$,

$$\dot{g} = -G_0^{-1}(e_{GA} - e_{AG}). \quad (26)$$

The limit $G_0 \rightarrow \infty$ freezes the donor coordinate but does not restore the working completion's derived target; it is therefore not a regular perturbation of the working vector field. If a sustained extraction flux satisfies $c(t) \geq c_0 > 0$ over an interval, the closed donor budget gives the order bound $T \leq G_0/c_0$. This is a budget bound, not an asymptotic estimate of delay-induced cycles. Whether a frozen-donor Hopf structure persists as a slowly drifting transient is an open slow-passage question.

S3. Full application tables and status records

S3.1 Classification matrix

Table 2: Three records and the time-like quantities they do or do not instantiate.

Quantity	Question	G3P	Phosphate	Fisheries
$J_A^{\text{gross}}, H_A^{\text{gross}}$	turnover/ dependency	no	no	gross-loss analogue only
H_A^{loc}	frozen net-rate ratio	no: anomaly, not stock	no: classification, not stock	no: no net $\dot{B}$
T_A	scenario hitting time	no	no	no
Object status	strongest permissible reading	record-relative statistical index	economic reserve-class quotient	removals-only pressure scale

S3.2 Application contract: observable, denominator and non-claim

The numeric tables are most safely read with the following contract. “Barrier” means the declared reference used by the arithmetic; it does not mean that the source has identified a physical failure floor.

Table 3: Application contract for the three public-data records.

Record	Observable	Unit	Denom.	Barrier/ reference	Interpretation	Source/ status	Explicit non-claim
G3P/ GRACE groundwa- ter	basin-mean anomaly a and fitted trend $\hat{a}$	cm; cm/yr	$[-\hat{a}]_+$	record- window minimum	persistence of an observed anomaly relative to its own window	G3P v1.12; GRACE- derived anomaly; basin extraction caveat	not absolute aquifer volume, recharge rate, physical depletion or forecast
USGS phosphate	reserve mass G_{reserve} and production C_G	kt; kt/yr	C_G	reserve classification	arithmetic reserve-life ratio	USGS MCS 2026 world pin; country rows retain vintage notes	not geological exhaustion; reserves are not resources and classification can change
RAM Legacy fisheries	$SSB_{\text{now}},$ B_{lim} and F_{now}	biomass; yr^{-1}	F_{now}	$B_{\text{lim}} =$ 0.2 max SSB	removals-only pressure time	archived 43-stock pull plus v4.44/v4.66 public com- parisons	not net biomass decline, demographic hitting time, policy forecast or stock-recruit model

S3.3 G3P groundwater anomaly table

For a basin anomaly $a(t)$, the reported index is

$$L_{\text{hist}}^{\text{anom}} = \frac{a_{\text{latest}} - a_{\text{hist,min}}}{[-\hat{a}]_+}. \quad (27)$$

The product supplies anomalies relative to a reference period, not absolute aquifer volumes. The window, basin mask, anomaly reference and fitted trend are part of the record status.

Table 4: G3P basin-row extraction used to exhibit the record-relative index.

Basin	Trend (cm/yr)	2023 anomaly (cm)	Implied window minimum (cm)	Index (yr)
Indo-Gangetic (N. India) ^a	−49.7	−414	−548	≈ 2.7
North China Plain	−18.6	−145	−292	≈ 7.9
Central Valley (US)	−16.1	−84	−237	≈ 9.5
La Mancha (Spain)	−3.2	−20	−88.5	≈ 21.4
High Plains (US)	−7.9	−160	−160	already at minimum
global mean	−0.4	−14	−33.0	≈ 47.5

^a The Indo-Gangetic magnitude is quarantined. It sits well beyond published basin-mean trends and must be re-derived from the product’s basin masks before numerical reuse. The classification status does not depend on the magnitude.

S3.4 Phosphate reserve table

The arithmetic reserve-life quotient is $T_{\text{reserve}} = G_{\text{reserve}}/C_G$. On the pinned record it is approximately 309 years for world reserves. It is an economic classification quotient, not a geological exhaustion forecast.

Table 5: Phosphate reserve-class table and arithmetic quotient.

Country/class	Reserves (kt)	Reserve-life horizon (yr)	Implied production (kt/yr)
China	3,400,000	≈ 28	121,429
United States	1,000,000	≈ 45	22,222
Jordan	820,000	≈ 62	13,226
Morocco	50,000,000	$\approx 1,250$	40,000
Australia ^a	5,800,000	$\approx 2,088$	2,778
World reserves	74,000,000	≈ 309	239,482
World resources, $\varepsilon = 0.10$	$> 300,000,000$	$> 1,125$	240,000

^a The Australia row carries a pre-2026 reserve vintage and is not used to establish the world classification. Reserve membership changes with prices, technology, exploration and regulation; a resources threshold is a different declared object.

S3.5 Fisheries removals-only table

For the declared comparison process $\dot{B} = -F_{\text{now}}B$, with $B_{\text{lim}} = 0.2 \text{ max SSB}$,

$$\Theta_F = \frac{\log(\text{SSB}_{\text{now}}/B_{\text{lim}})}{F_{\text{now}}}. \quad (28)$$

It omits recruitment, growth, maturation, natural mortality, density dependence, environmental forcing and future policy.

Table 6: Fisheries cohort and release sensitivity record.

Record	Stocks	Median Θ_F (yr)	Status and caveat
Archived selected cohort	43	≈ 1.8	Headline cohort; eight zero entries at or below reference; not reproduced by either public release
Archived positive sub-cohort	35	≈ 2.9	$F > 0$ and $\text{SSB}_{\text{now}} > B_{\text{lim}}$; still removals-only

Record	Stocks	Median Θ_F (yr)	Status and caveat
RAM Legacy v4.44 public release	415	≈ 2.57	Same protocol; public-release comparison, not the archived cohort
RAM Legacy v4.66 public release	454	≈ 3.39	Same protocol; version-sensitive broad cohort

The selected 43-stock record is a class-specific diagnostic rather than a statistic of all assessed fisheries. The broad release comparison is a sensitivity record, not an uncertainty interval around a demographic forecast.

S4. Operational weak and strong sustainability regimes

S4.1 Declared closure objects

Let $\mathcal{V}(x, t)$ be the set of non-negative primitive flux vectors satisfying the typed incidence rules, donor limits, conversion declarations and boundary constraints at (x, t) . Let $D(t)$ be a declared service-demand vector and P map return fluxes to the uses they cover. Let $m(x) = Gx + a$ be the declared componentwise margin. Define the closure capacity

$$\Lambda^*(x, t) = \sup \{ \lambda \geq 0 : \exists v \in \mathcal{V}(x, t), \quad \lambda D(t) \leq Pv, \quad m(x) \geq 0 \}. \quad (29)$$

The inequality in (29) is a declared coverage relation; it does not equate a service unit with a material unit. Cross-type substitution enters only through a declared conversion column and its coefficient. An external input is a boundary input, not internal regeneration.

For a horizon T , let $\mathcal{A}_T$ be the set of admissible trajectories satisfying the typed balance, donor and conversion rules, declared boundary schedule and margin constraints on $[0, T]$. The closure object is therefore a feasible-set statement, not a scalar sustainability score.

S4.2 Definitions

Definition S4.1 (Operational weak-closure regime). The declared ledger is in the weak-closure regime on $[0, T]$ if there exists a trajectory in $\mathcal{A}_T$ that meets the declared demand and whose return network has $\Lambda^*(x(t), t) \geq 1$ at the relevant use times. Declared substitutions are admissible only when their type conversion, coefficient, destination and residual route are present.

Definition S4.2 (Operational strong-component regime). The declared ledger is in the strong-component regime on $[0, T]$ when a critical component or barrier is a live certification obligation and either (i) no admissible conversion/return route covers its demand at the declared horizon, $\Lambda^* < 1$, or (ii) a componentwise margin is binding or violated. A surplus in another component does not clear that obligation.

These definitions are operational readings of a typed ledger. They do not assert that all ethical, distributive or intergenerational meanings of weak and strong sustainability reduce to Λ^* and m .

S4.3 Conditional regime proposition

Proposition S4.3 (What the regime labels certify). If the weak-closure conditions hold, the declared return network can cover the declared demand on the stated horizon while satisfying the declared componentwise margins. If the strong-component condition holds, the record identifies a binding component, a deficit in declared return capacity, or an unestablished route; no non-negative scalar aggregate certifies the missing component without an additional domain implication.

Proof. The first statement is the feasibility witness in (29) together with membership in $\mathcal{A}_T$. For the second, either $\Lambda^* < 1$ by definition or a live componentwise margin fails. A scalar aggregate that remains positive while a component is negative is not equivalent to the conjunction of component inequalities; the counterexample $b = (-2, 3)$ with $w = (1/2, 1/2)$ supplies the witness. $\square$

S4.4 Formalization boundary

A universal weak/strong-sustainability theorem would require a complete class of closure maps, admissible substitutions, boundary flows, time horizons, critical-stock definitions and institutional valuation rules. Those objects are not supplied by a generic MFA table. Accordingly, this supplement formalizes only the operational regime readout in Definitions S4.1–S4.2. The safe claim is narrower: the ledger makes the physical substitution and closure assumptions explicit, and it refuses cross-type compensation as a certification shortcut.

Table 7: Reporting template for the two operational regimes.

Field	Weak-closure reading	Strong-component reading
Physical route	Declared conversions and return routes cover demand	Route absent, insufficient or not identified
Closure capacity	$\Lambda^* \geq 1$ at the declared use times	$\Lambda^* < 1$ or not established
Component margin	$m_j(t) \geq 0$ on the declared horizon	A critical m_j binds, fails or is not established
Scalar aggregate	Communication or ranking only	Cannot clear the component obligation
Status wording	State the witness and horizon	Name the deficit, missing route or binding barrier

S5. Reproducibility and non-claims

The application tables are source-vintage records and bounded arithmetic. The delay section specifies an interface and a non-reduction boundary; results for the open institutional completion are not assigned to the closed ledger.

The principal article record is available at <https://doi.org/10.6084/m9.figshare.33942451>.

References

- Abaee, A., 2026. *Typed Flux Ledgers and Depletion Arithmetic: Conservation, Componentwise Diagnostics, and the Semantics of Depletion Horizons*. Figshare preprint. <https://doi.org/10.6084/m9.figshare.33942451>.
- Abaee, A., 2026a. *Certifying a Typed Ledger: The Predicates, the Programmes, the Vintages, and the Reproduction Bundle*. Figshare preprint. <https://doi.org/10.6084/m9.figshare.33942469>.
- Abaee, A., 2026b. *What the Accounts Settle, and What They Leave Open: Depletion as a Cost of Production, and the Missing Step from a Rate to a Horizon*. Figshare preprint. <https://doi.org/10.6084/m9.figshare.33942487>.
- Ahuja, R.K., Magnanti, T.L., Orlin, J.B., 1993. *Network Flows: Theory, Algorithms, and Applications*. Prentice-Hall.
- Aubin, J.-P., 1991. *Viability Theory*. Birkhäuser.
- Brunner, P.H., Rechberger, H., 2004. *Practical Handbook of Material Flow Analysis*. Lewis Publishers.
- Chhikara, R.S., Folks, J.L., 1989. *The Inverse Gaussian Distribution*. Marcel Dekker.
- Daly, H.E., 1990. Toward some operational principles of sustainable development. *Ecological Economics* 2, 1–6.
- Ekins, P., Simon, S., Deutsch, L., Folke, C., De Groot, R., 2003. A framework for the practical application of the concepts of critical natural capital and strong sustainability. *Ecological Economics* 44, 165–185.
- Feinberg, M., 2019. *Foundations of Chemical Reaction Network Theory*. Springer.
- Güntner, A., Sharifi, E., Haas, J., et al., 2024. Global Gravity-based Groundwater Product (G3P), v1.12. GFZ Data Services. <https://doi.org/10.5880/G3P.2024.001>.
- Jacquez, J.A., Simon, C.P., 1993. Qualitative theory of compartmental systems. *SIAM Review* 35, 43–79.
- Munda, G., Nardo, M., 2009. Noncompensatory/nonlinear composite indicators for ranking countries: a defensible setting. *Applied Economics* 41, 1513–1523.
- Neumayer, E., 2013. *Weak versus Strong Sustainability: Exploring the Limits of Two Opposing Paradigms*, 4th ed. Edward Elgar.
- Oksendal, B., 2003. *Stochastic Differential Equations*, 6th ed. Springer.
- Redner, S., 2001. *A Guide to First-Passage Processes*. Cambridge University Press.
- Ricard, D., Minto, C., Jensen, O.P., Baum, J.K., 2012. Examining the knowledge base and status of commercially exploited marine species with the RAM Legacy Stock Assessment Database. *Fish and Fisheries* 13, 380–398.
- Tilton, J.E., 2003. *On Borrowed Time? Assessing the Threat of Mineral Depletion*. Resources for the Future.
- U.S. Geological Survey, 2026. *Mineral Commodity Summaries 2026: Phosphate Rock*. USGS.
- Abaee, A., 2026c. *Institutional delay dynamics companion analysis*. Zenodo. <https://doi.org/10.5281/zenodo.22554217>.